

AKANKSHA RAI

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EDUCATION

UNIVERSITY OF MICHIGAN ANN-ARBOR

Ann Arbor, MI

Bachelor of Science in Computer Science

Expected May 2026

Relevant Coursework: Data Structures and Algorithms, Machine Architecture; Practical Data Science; Program

Design and Development, Foundations of Computer Science

UNIVERSITY OF MINNESOTA TWIN CITIES

Minneapolis, MN

Cumulative GPA: 3.96/4.0 ; Dean's List: 2022-2024

Fall 2022 - Spring 2024

TECHNICAL SKILLS

Languages: Python, Java, HTML, CSS, Javascript, R, C, Assembly, C++, OCaml

Tools/Platforms: Git, Github, Figma, VSCode, Node.js, AndroidStudio, Zap, MySQL, Docker, MongoDB

Frameworks: Flask, NumPy, Pandas, RestAPIs, React, Robot Framework, Jenkins, Typescript

EXPERIENCE

Software Engineering Intern – Application Security

New York, NY

MongoDB

Jun 2025 – Aug 2025

- Engineered a scalable GitHub auditing framework in Python, leveraging GitHub REST and GraphQL APIs, to automatically detect security misconfigurations, and ensure compliance with AppSec policies and internal security baselines
- Developed a full-stack MERN (MongoDB, Express.js, React, Node.js) web dashboard for real-time visualization and executive reporting of repository audit results, improving visibility for security teams and leadership.
- Performed manual and automated web application penetration testing on internal tools (e.g., VoyageAI), utilizing Burp Suite, OWASP Top 10, and custom Python scripts to identify and triage vulnerabilities. Provided detailed remediation plans to engineering teams, aligning fixes with secure SDLC practices.

Teaching Assistant - Applied Data Science/Machine Learning

Ann Arbor, MI

University of Michigan

Jan 2025 – May 2025

- Led weekly discussion sessions and office hours for 50+ students, guiding them through data manipulation libraries (numpy, pandas), strengthening hands-on coding skills and debugging practices.
- Assisted in assignments spanning API integration (web scraping with BeautifulSoup), data wrangling, statistical modeling (linear/logistic regression, gradient descent), cross-validation, and unsupervised learning—covering the full data science pipeline

Cybersecurity Software Engineering Intern

Minneapolis, MN

Trane Technologies

May 2024 – Aug 2024

- Automated active scans through OWASP's ZAP Tool using Python's Robot Framework to automatically spider through websites to identify potential security vulnerabilities resulting in reduced test result time by 66%
- Enhanced website's reliability by implementing a CI/CD pipeline with Jenkins to continuously monitor and alert architects on possible security issues using JIRA's REST API to automatically create tickets

PROJECTS

Drone Delivery Simulation

Jan 2024 - April 2024

- Developed key features for a drone simulation using Observer and Factory design patterns, such as battery charging systems that dynamically create and manage charging stations and implementing thief protection of packages utilizing data analysis.
- Engaged in a waterfall/test-driven development lifecycle, executing software development activities including writing requirements, designing UMLs, coding, testing, and documenting code following Google documentation conventions.

GreenChoice - Google Solutions Challenge

Nov 2023 - Feb 2024

- Developed a mobile application using AndroidStudio to promote the switch to sustainable, greener alternatives in individuals' daily lives. Created a logging system using Java and Firebase to track "green choices" made daily, and also provided educational content and an interactive user interface.

Validify - 1st Place in University of Minnesota Makeathon

March 2024

- Leveraged AWS's Bedrock tool to ideate and develop an extension to combat misinformation on social media, utilizing credible medical sources to train a model that generates trust percentages for posts, enhancing transparency and accuracy in information dissemination

Society of Women Engineers Team Tech - John Deere

March 2024

- Collaborated with John Deere's engineering and product teams to understand pain points and requirements, gained first-hand experience in systems engineering lifecycle tailored to meet needs of industry clients